

B2B Multi-Vendor Marketplace System

1. Introduction

A B2B Multi-Vendor Marketplace is a platform where multiple vendors sell products or services to businesses. It allows bulk purchasing, vendor management, and centralized control through an admin panel.

2. Objectives

- 1 Provide a platform for multiple vendors to sell products
- 2 Enable businesses to purchase in bulk
- 3 Centralized management for admin
- 4 Secure and scalable architecture

3. Key Features

Vendor Panel

- 1 Product management
- 2 Inventory control
- 3 Order management
- 4 Analytics dashboard

Buyer Panel

- 1 Browse products
- 2 Bulk order system
- 3 Order tracking
- 4 Secure payments

Admin Panel

- 1 User management
- 2 Vendor approval
- 3 Commission management
- 4 Reports and analytics

4. System Architecture

Frontend: React + Tailwind CSS Backend: Node.js + Express Database: MongoDB Authentication: JWT

5. Database Design

- 1 Users Collection (Admin, Vendor, Buyer)
- 2 Products Collection
- 3 Orders Collection
- 4 Payments Collection
- 5 Vendor Profiles

6. Workflow

- 1 User Registration/Login
- 2 Vendor adds products
- 3 Buyer places order
- 4 Payment processing
- 5 Order fulfillment

7. Future Enhancements

- 1 AI-based recommendations
- 2 Real-time chat between buyer and vendor
- 3 Mobile application
- 4 Advanced analytics